

Safety Investigation Report

Loss of separation in Sector E3 after a coasting radar track (illustrative)

Report	NAVRAP-INV-OCC-RDR-01
Project	DEMO-08 — Surveillance radar performance degradation
Status	Example (illustrative, not a real occurrence)
Prepared by	Safety & Quality — illustrative NAVRAP worked example
Source	Illustrative occurrence for NAVRAP DEMO-08
Generated	2026-10-02

NAVRAP inv-report-1.0.0 · analysis engine atsb-1.1.0

1. Executive summary

1.1 Synopsis

What	Loss of separation in Sector E3 after a coasting radar track (illustrative)
Occurrence type	Loss of separation (serious incident)
When	2026-08-21 14:07 LT
Where	ACC Sector E3 (FL100–FL245)
Who	AirNav (ATS, CNS)
Reference	OCC-RDR-01

A coasting track hid the converging AAA101 from the E3 controller, who climbed BBB202 to FL190. STCA alerted late and avoiding action restored separation at 3.1 NM.

1.2 Consequences

No injuries or damage. Serious incident: separation 3.1 NM / 0 ft against 5 NM required. No environmental impact.

1.3 Core finding

The ORLIO analysis identified 8 contributing safety factors and 1 other factors that increased risk. 2 safety issues were identified (2 significant). The primary safety breakdowns, traced from the occurrence back to the organisational level, were:

- Occurrence: Separation between AAA101 and BBB202 reduced to 3.1 NM at FL190 where 5 NM was required; The track of AAA101 coasted for three scans and the tracker's extrapolation diverged from the aircraft's turn.
- Individual actions: The controller planned the climb of BBB202 on the displayed (coasted) position of AAA101.
- Local conditions: The radar's probability of detection between 030° and 090° had degraded to about 94 %; AAA101 was not ADS-B equipped, so no second source could correct the coasted track.
- Risk controls that were absent, failed or were bypassed: Radar monitoring raised alarms for hard failures only; there were no thresholds for detection probability or azimuth bias; No procedure defined criteria for declaring degraded surveillance or the separation to apply; The coasting symbol was not visually distinct from an updated track.

Findings are stated to identify safety factors, not to apportion blame or liability.

2. Factual information

2.1 Occurrence timeline

Time	Event	Source
2026-06-01	Monthly flight-check data first show Pd below 98 % in the 030°–090° sector; no threshold exists, no action	Performance data (retrospective)
2026-07-15	A telecommunication tower 1.2 km north-east of the radar site is commissioned	Site survey
2026-08-21 14:02	AAA101 (not ADS-B equipped) enters E3 at FL190 north-east bound	SDPS recording
2026-08-21 14:05	The track of AAA101 coasts for three scans (36 s) and drifts from the aircraft's actual turn	SDPS recording; radar plot data
2026-08-21 14:05	The controller plans BBB202's climb to FL190 behind the displayed position of AAA101	RT recording; interview
2026-08-21 14:06	The track of AAA101 updates and jumps 1.8 NM toward BBB202	SDPS recording
2026-08-21 14:06	STCA alerts 32 s before the closest point of approach	System log
2026-08-21 14:06	The controller instructs BBB202 to turn right 40 degrees, avoiding action	RT recording
2026-08-21 14:07	Separation reduces to 3.1 NM at FL190 (5 NM required)	Radar replay
2026-08-21 14:20	The controller files an occurrence report; the supervisor informs the ATSEP	Occurrence report
2026-08-25	Inspection finds the rotary joint worn beyond its hours limit and encoder bearing play	Maintenance report

2.2 Personnel and assets involved

Personnel	Experience, qualifications and duty
E3 executive controller	9 years' experience; valid rating; second hour of an afternoon shift
ACC supervisor	Aware of informal reports of 'jumpy tracks'; no declaration in force
ATSEP on duty	Daily checks completed; no RMM alarm

Asset / equipment	Type and configuration
Bukit Sari MSSR Mode S	Mode S MSSR, 12 s scan, single radar coverage below FL150 in the NE quadrant; rotary joint beyond hours limit
SDPS tracker	Multi-sensor tracker; coasting up to 5 scans
Aircraft	AAA101 (not ADS-B equipped) and BBB202 (ADS-B equipped), both at FL190 after the climb

Environmental configuration	Daytime, visual meteorological conditions; moderate traffic (14 flights in E3).
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2.3 Immediate actions taken

- Avoiding action to BBB202; traffic information to both aircraft
- Supervisor applied 10 NM in the NE quadrant until the end of the day (informal)
- SDPS and radar plot recordings secured; ATSEP performance analysis requested

3. ORLIO analysis framework

Evidence is mapped against the ORLIO layers to trace the causal flow from the occurrence back to organisational influences. Each factor was tested for existence, and for influence on the occurrence or on another factor (or for importance), to the standard of proof 'likely' (≥66 %), following the ATSB safety investigation analysis method.

3.1 Occurrence events (O)

Definition: The final physical outcomes and the specific technical failure mechanisms that triggered the accident.

Control of the separation was lost when the coasted track of AAA101 updated 1.8 NM from where it was displayed.

F1 · Separation between AAA101 and BBB202 reduced to 3.1 NM at FL190 where 5 NM was required

Classification	Occurrence event
Finding	Contributing safety factor
Existence	supported (Virtually certain)
Influence	supported (Virtually certain) — on the occurrence: the occurrence
Explained by	F3 The controller planned the climb of BBB202 on the displayed (coasted) position of AAA101

- [supports] Radar replay and ADS-B data of BBB202 give the closest point of approach — Radar replay
- [supports] The infringement is the occurrence — Radar replay

F2 · The track of AAA101 coasted for three scans and the tracker's extrapolation diverged from the aircraft's turn

Classification	Technical failure mechanism · T2
Finding	Contributing safety factor
Description	Plot data show missed detections of AAA101 in three consecutive scans at azimuth 052°; the rotary joint was later found worn.
Existence	supported (Very likely)
Influence	supported (Very likely) — on F3 (Individual action): The controller planned the climb of BBB202 on the displayed (coasted) position of AAA101
Explained by	F4 The radar's probability of detection between 030° and 090° had degraded to about 94 %, F5 AAA101 was not ADS-B equipped, so no second source could correct the coasted track

- [supports] Plot data: no plots for AAA101 at 14:05:12, 14:05:24, 14:05:36 — Radar plot data
- [supports] Rotary joint worn beyond hours limit — Maintenance report 25 August
- [supports] The displayed position was 1.8 NM from the actual position when the plan was made — Radar replay vs ADS-B of BBB202 and FDR of AAA101

3.2 Individual actions (I)

Definition: The direct actions, omissions or decisions made by frontline operators (pilots, controllers, drivers, technicians).

The plan was correct for the displayed picture; the display was wrong.

F3 · The controller planned the climb of BBB202 on the displayed (coasted) position of AAA101

Classification	Individual action · I3.1
Finding	Contributing safety factor
Role / error type	Air traffic controller · information
Why it made sense at the time	The coasting symbol differs from a normal track only by a small mark; tracks had been 'jumpy' for weeks and usually recovered; no degraded mode was in force, so 5 NM on the display was the normal practice.
Existence	supported (Virtually certain)
Influence	supported (Very likely) — on F1 (Occurrence event): Separation between AAA101 and BBB202 reduced to 3.1 NM at FL190 where 5 NM was required
Explained by	F2 The track of AAA101 coasted for three scans and the tracker's extrapolation diverged from the aircraft's turn, F7 No procedure defined criteria for declaring degraded surveillance or the separation to apply, F8 The coasting symbol was not visually distinct from an updated track

- [supports] Climb clearance at 14:05:40 with 5.5 NM displayed spacing — RT recording; radar replay
- [supports] Controller interview — Interview
- [supports] With the true position the climb would not have been issued — Radar replay

F11 · The controller gave avoiding action immediately after the STCA alert

Classification	Positive action · I3.3
Finding	Positive factor
Role / error type	Air traffic controller · action
Why it made sense at the time	Trained response to STCA.
Existence	supported (Virtually certain)
Influence	supported (Very likely) — on the occurrence: the occurrence
Explained by	F10 STCA alerted 32 seconds before the closest point of approach

- [supports] RT recording 14:06:40 — RT recording
- [supports] Without the turn the aircraft would have passed about 1 NM apart — Radar replay extrapolation

3.3 Local conditions (L)

Definition: Environmental, situational or task-specific variables affecting human performance at the time of the event.

Detection probability had degraded in the very sector where AAA101 flew, and no second sensor covered it.

F4 · The radar's probability of detection between 030° and 090° had degraded to about 94 %

Classification	Local condition · L6.3
Finding	Contributing safety factor
Existence	supported (Virtually certain)
Influence	supported (Likely) — on F2 (Technical failure): The track of AAA101 coasted for three scans and the tracker's extrapolation diverged from the aircraft's turn
Explained by	F6 Radar monitoring raised alarms for hard failures only; there were no thresholds for detection probability or azimuth bias
Further explanation	Explained by the worn rotary joint (F2 evidence) and by the missing performance thresholds (F6).

- [supports] Performance analysis of August data: Pd 94.1 % in 030°–090° — Performance report
- [supports] Lower Pd makes multi-scan coasting more likely — Radar specialist

F5 · AAA101 was not ADS-B equipped, so no second source could correct the coasted track

Classification	Local condition · L3.6
Finding	Contributing safety factor
Existence	supported (Virtually certain)
Influence	supported (Likely) — on F2 (Technical failure): The track of AAA101 coasted for three scans and the tracker's extrapolation diverged from the aircraft's turn
Further explanation	ADS-B carriage below FL245 is not mandated; outside AirNav's control.

- [supports] Flight plan equipment field and SDPS sensor data — Flight plan; SDPS
- [supports] ADS-B updates would have kept the fused track on the aircraft — SDPS specialist

3.4 Risk controls (R)

Definition: Existing safeguards, defences, barriers or safety procedures meant to intercept errors or break the accident chain.

Monitoring, procedure and HMI defences were all absent or weak; only STCA and the controller's response worked.

F6 · Radar monitoring raised alarms for hard failures only; there were no thresholds for detection probability or azimuth bias

Classification	Risk control · R6.3, R1.4
Finding	Contributing safety factor
Control function	preventive
Existence	supported (Virtually certain)
Influence	supported (Very likely) — on F4 (Local condition): The radar's probability of detection between 030° and 090° had degraded to about 94 %
Explained by	F9 Controllers reported track anomalies informally rather than through the occurrence system
Safety issue	Tolerable (upper) — risk 3B (AirNav 5×5); owner CNS engineering

- [supports] RMM configuration: no performance thresholds — RMM configuration
- [supports] Maintenance manual: performance analysis 'as required' — Maintenance manual
- [supports] A threshold at 97 % would have flagged the June data — Performance data (retrospective)

F7 · No procedure defined criteria for declaring degraded surveillance or the separation to apply

Classification	Risk control · R3
Finding	Contributing safety factor
Control function	preventive
Existence	supported (Virtually certain)
Influence	supported (Likely) — on F3 (Individual action): The controller planned the climb of BBB202 on the displayed (coasted) position of AAA101
Further explanation	The Manual of ATS was written for total radar failure; degraded performance was never assessed as a hazard (see F6).
Safety issue	Tolerable (upper) — risk 3B (AirNav 5×5); owner ATS standards

- [supports] Manual of ATS covers total radar failure only — Manual of ATS §8
- [supports] With a declaration, 8 NM would have applied in the NE quadrant — Investigation team

F8 · The coasting symbol was not visually distinct from an updated track

Classification	Risk control · R1.1
Finding	Contributing safety factor
Control function	recovery
Existence	supported (Virtually certain)
Influence	supported (Likely) — on F3 (Individual action): The controller planned the climb of BBB202 on the displayed (coasted) position of AAA101
Further explanation	HMI specification inherited from the 2012 system; no human-factors review of coasting presentation.

- [supports] HMI specification: coasting shown by a 2-pixel mark — HMI specification
- [supports] Controller did not notice the coasting — Interview

F10 · STCA alerted 32 seconds before the closest point of approach

Classification	Positive condition · R1.4
Finding	Positive factor
Existence	supported (Virtually certain)
Influence	supported (Very likely) — on F11 (Positive action): The controller gave avoiding action immediately after the STCA alert

- [supports] System log — STCA log
- [supports] The alert prompted the avoiding action — RT recording; interview

3.5 Organisational influences (O)

Definition: Latent systemic, corporate or policy-level choices that fostered unsafe local conditions or weakened risk controls.

The maintenance regime and the reporting culture did not turn weeks of visible degradation into action.

F9 · Controllers reported track anomalies informally rather than through the occurrence system

Classification	Organisational influence · O1
Finding	Other factor that increased risk
Existence	supported (Very likely)
Influence	not_supported (More likely than not) — on F6 (Risk control): Radar monitoring raised alarms for hard failures only; there were no thresholds for detection probability or azimuth bias
Importance	Reporting is how controller-visible degradation reaches maintenance; worth addressing.

- [supports] Nine formal reports June–August vs many verbal complaints — Occurrence database; interviews
- [supports] More reports might have triggered an earlier inspection — Investigation team

4. Findings and contributing factors

Contributing safety factors are listed in priority order — safety issues by risk level first, then from the organisational level down to the occurrence. Each entry shows the chain through which it contributed (→ influenced).

4.1 Contributing factors (chain of events and conditions)

Occurrence: F1 Separation between AAA101 and BBB202 reduced to 3.1 NM at FL190 where 5 NM was required; F2 The track of AAA101 coasted for three scans and the tracker's extrapolation diverged from the aircraft's turn.

- 1 Contributing factor 1 — Risk control: Radar monitoring raised alarms for hard failures only; there were no thresholds for detection probability or azimuth bias. [Safety issue · Tolerable (upper) 3B] Chain: F6 (Risk control) → F4 (Local condition) → F2 (Technical failure) → F3 (Individual action) → F1 (Occurrence event) → the occurrence.
- 2 Contributing factor 2 — Risk control: No procedure defined criteria for declaring degraded surveillance or the separation to apply. [Safety issue · Tolerable (upper) 3B] Chain: F7 (Risk control) → F3 (Individual action) → F1 (Occurrence event) → the occurrence.
- 3 Contributing factor 3 — Risk control: The coasting symbol was not visually distinct from an updated track. Chain: F8 (Risk control) → F3 (Individual action) → F1 (Occurrence event) → the occurrence.
- 4 Contributing factor 4 — Local condition: The radar's probability of detection between 030° and 090° had degraded to about 94 %. Chain: F4 (Local condition) → F2 (Technical failure) → F3 (Individual action) → F1 (Occurrence event) → the occurrence.

- 5 Contributing factor 5 — Local condition: AAA101 was not ADS-B equipped, so no second source could correct the coasted track. Chain: F5 (Local condition) → F2 (Technical failure) → F3 (Individual action) → F1 (Occurrence event) → the occurrence.
- 6 Contributing factor 6 — Individual action: The controller planned the climb of BBB202 on the displayed (coasted) position of AAA101. Chain: F3 (Individual action) → F1 (Occurrence event) → the occurrence.
- 7 Contributing factor 7 — Technical failure: The track of AAA101 coasted for three scans and the tracker's extrapolation diverged from the aircraft's turn. Chain: F2 (Technical failure) → F3 (Individual action) → F1 (Occurrence event) → the occurrence.

4.2 Other factors that increased risk

- 1 Organisational influence: Controllers reported track anomalies informally rather than through the occurrence system.

4.3 Positive factors

- STCA alerted 32 seconds before the closest point of approach
- The controller gave avoiding action immediately after the STCA alert

4.4 Other key findings

None.

4.5 Safety issues

Ref	Safety issue	Owner	Risk	Status
F6	Radar monitoring raised alarms for hard failures only; there were no thresholds for detection probability or azimuth bias	CNS engineering	3B Tolerable (upper)	Safety action pending
F7	No procedure defined criteria for declaring degraded surveillance or the separation to apply	ATS standards	3B Tolerable (upper)	Safety action pending

4.6 Factors considered and not included

Ref	Factor considered	Reason
F12	The controller lost situational awareness	Restates the problem (e.g. 'loss of situation awareness') — Restates F3; the reasons are F2, F4–F8.

5. Safety recommendations and corrective actions

Every identified contributing factor is resolved with corrective action, ranked by the hierarchy of controls: Elimination > Substitution > Engineering controls > Administrative controls > Personal protective equipment. Formal safety recommendations state the safety issue; the addressee determines the means. [Date] marks a target date still to be agreed.

ORLIO layer	Identified deficiency	Corrective action required	Action owner	Target date
Organisational	F9 Controllers reported track anomalies informally rather than through the occurrence system	[Administrative controls] Track-anomaly report category with feedback to ATSEP; SPI S-05 (proposed (NAVRAP example); Proposed)	ACC unit chief	[Date]
Risk controls	F6 Radar monitoring raised alarms for hard failures only; there were no thresholds for detection probability or azimuth bias	[Engineering controls] Monthly radar performance analysis with thresholds (Pd ≥ 97 % per 60° sector, bias ≤ 0.1°, false targets) and automatic RMM alerts (proposed (NAVRAP example); Proposed)	CNS engineering	2026-12-31
Risk controls	F6 Radar monitoring raised alarms for hard failures only; there were no thresholds for detection probability or azimuth bias	[Engineering controls] Hours-based replacement of rotary joints and encoders; critical spares (proposed (NAVRAP example); Proposed)	Head of CNS maintenance	[Date]
Risk controls	F7 No procedure defined criteria for declaring degraded surveillance or the separation to apply	[Administrative controls] Degraded-surveillance procedure with criteria, standard declaration form, CWP status banner and 8 NM minimum (proposed (NAVRAP example); Proposed)	ATS standards	2026-11-30
Risk controls	F8 The coasting symbol was not visually distinct from an updated track	[Engineering controls] Distinct coasting symbol and age-of-update indication (proposed (NAVRAP example); Proposed)	ATM systems	[Date]

ORLIO layer	Identified deficiency	Corrective action required	Action owner	Target date
Local conditions	F4 The radar's probability of detection between 030° and 090° had degraded to about 94 %	[Administrative controls] Declare degraded mode whenever Pd < 97 % in any 60° sector until repaired (see A1, A3) (proposed (NAVRAP example); Proposed)	CNS engineering	[Date]
Local conditions	F5 AAA101 was not ADS-B equipped, so no second source could correct the coasted track	[Administrative controls] Safety recommendation: Consider an ADS-B carriage requirement below FL245 in the eastern FIR (NAVRAP proposal to the regulator; Proposed)	DGCA	[Date]
Individual actions	F3 The controller planned the climb of BBB202 on the displayed (coasted) position of AAA101	[Administrative controls] Briefing and recurrent training: verify the position of non-ADS-B aircraft by RT when a track coasts before using it for separation (proposed (NAVRAP example); Proposed)	ACC unit chief	[Date]
Occurrence events	F2 The track of AAA101 coasted for three scans and the tracker's extrapolation diverged from the aircraft's turn	[Elimination] Replace the rotary joint and azimuth encoder; re-align and verify registration (safety action taken; Closed)	CNS engineering	2026-10-12

Actions by hierarchy level: Elimination 1, Engineering controls 3, Administrative controls 5. Every contributing factor has at least one corrective action.

6. Appendices and supporting evidence

Appendix A — Photographs and diagrams of the scene and asset damage

No photographs or diagrams attached.

Appendix B — Interview summaries and witness accounts

E3 controller	Tracks had been jumpy for weeks; usually they recovered. Did not see the coasting mark.
ATSEP	No alarms; daily checks normal; performance analysis not routinely done.

Appendix C — ORLIO factor map

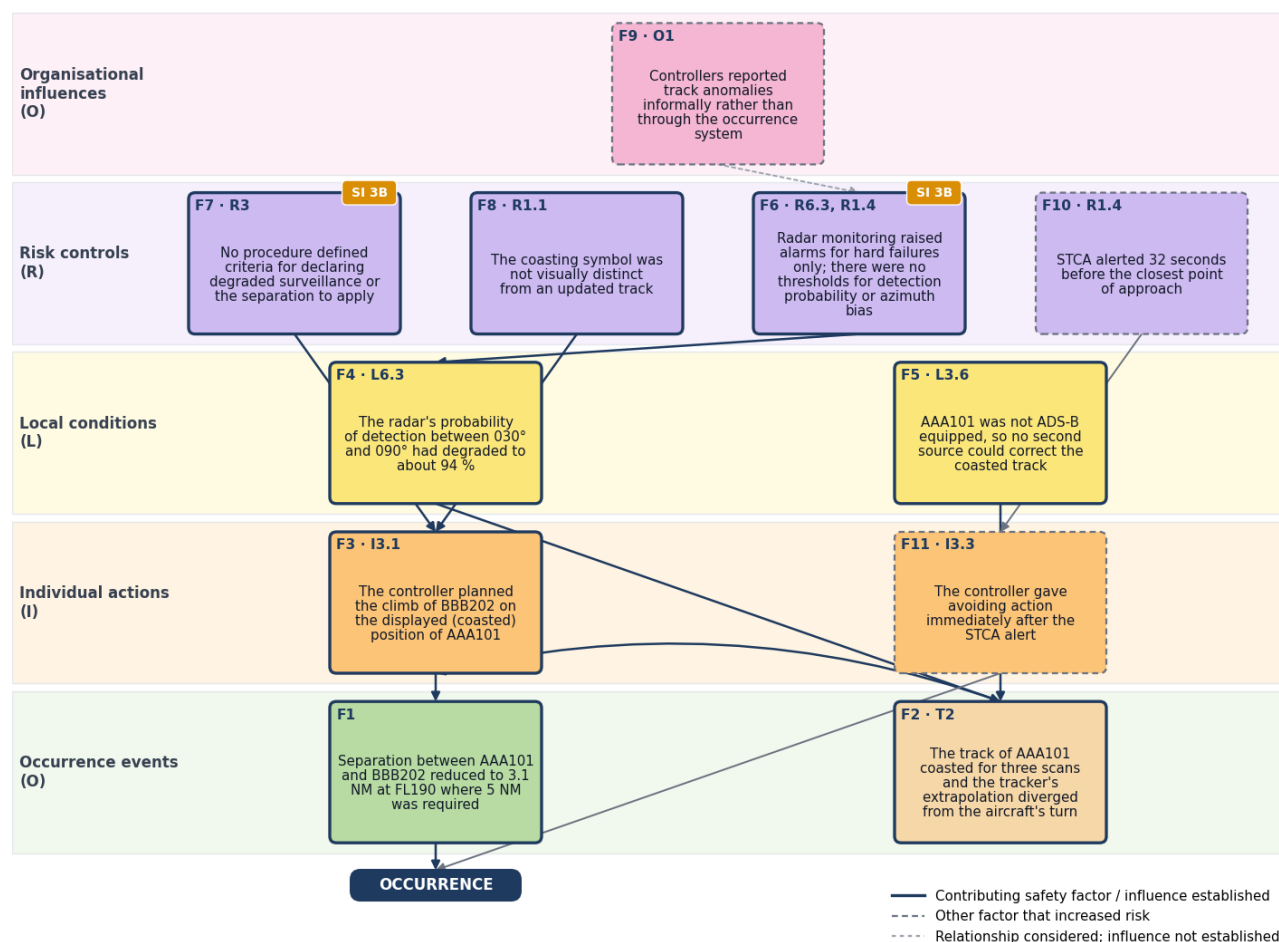


Figure C1. ORLIO factor map — solid borders are contributing safety factors, dashed borders other factors that increased risk; solid arrows show an established influence, dotted arrows a relationship considered but not established; SI badges mark safety issues with their risk index.